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Valence First?

Evaluating the Prominence of Valence with respect to truth-conditional meaning



VALENCE
ASYMMETRIES



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INTRODUCTION

What is valence?

Valence refers to the positive (pleasant) or negative (unpleasant) value attached to information.

The role of valence downplayed in the “truth-conditional approach »

▪ The truth-conditional approach

< work of Gottlob Frege (1892).

= centered around the referential relationship between words and the world.

- Meaning of a sentence = its truth conditions
= how the world should be for the sentence to be true.
(see, e.g., Davidson 1967; Montague 1970, 1973; Tarski 1956).

→ Valence has little to no place.

Ex : “M. Turner killed M. Smith”.

> No change with the “dynamic turn” (e.g., Kamp, 1981; Kamp & Reyle, 1979; Stalnaker, 1979).

> cf efforts to incorporate analyses of certain aspects of non-truth-conditional meaning in the 2000s (e.g., Potts, 2005, 2006; McCready, 2010, 2014; Gutzman, 2015; Jeshion, 2021).

Valence as a main object of inquiry in psychology

Osgood et al. (1957)

- They developed a measurement called the “semantic differential”.
= participants explicitly rate antonym word pairs wrt several different meaning dimensions.

- They identify three main dimensions of affective meaning :
 - valence
 - arousal
 - dominance.

- Other researchers have used multidimensional analyses to look at :
 - general stimuli (e.g., Mehrabian & Russel 1974; Russel et al 1989)
 - specifically linguistic stimuli (e.g., Bradley & Lang 1999; Warriner et al 2013; Mohammad 2018).

A novel experimental question

If valence is part of lexical meaning,

what is the respective significance of valence vs. truth-conditional content

in the processing of lexical meaning?

Possible views

- **The Valence-First Hypothesis (VFH)**

= valence information comes to mind prior to all other forms of content.

- **The Truth-Conditional First Hypothesis (TCFH)**

= truth-conditional content must be established first in order for valence information to attach to it.

- **The Valence Prominence Hypothesis (VPH)**

= valence may be more prominent than most – but not necessarily all – features of truth-conditional content.

The Valence-First Hypothesis (VFH)

Some indirect evidence from existing literature ?

- The “affective primacy hypothesis” (Zajonc 1980, 2000):
= affective reactions are the first that organisms have to stimuli.

Evidence that affective responses are activated quickly and automatically in the absence of explicit cognition:

- > **‘mere exposure’ paradigm:** repeated exposure to a novel stimulus increases subsequent preference for it (e.g., Zajonc, 1968, 1980, 2001; Monahan, Murphy, & Zajonc, 2000).
- > **evaluative and masked priming studies:** valence congruence between primes and targets reliably facilitates response times (e.g., Fazio et al., 1986; Murphy & Zajonc, 1993; Bargh et al., 1989; Wentura & Degner, 2010).
- > **comparison between ‘emotion words’ and neutral words:** emotion words are
 - pronounced faster (e.g., Bargh et al. 1996; Kissler et al. 2007),
 - remembered better (e.g., Herbert et al. 2006; Kensinger & Corkin 2003),
 - identified more rapidly than neutral words (e.g., Kousta et al. 2009).

The Truth-Conditional First Hypothesis (TCFH)

Some indirect evidence from existing literature ?

- Cognitive appraisal theories of emotion (e.g., Lazarus 1984) : emotion depends on first determining goal relevance.
- Rolls (1999) : reward/punishment can only be computed after perceptual recognition and object identification.
- Semantic priming is more robust, automatic, and seems to mediate affective priming (Storbeck, Robinson & McCour 2006; Storbeck & Clore 2007; Winkelman et al 2005).
- Early perceptual and semantic encoding processes precede affect responses in ERP studies (Calvo & Nummenmaa 2007; Batty, Cave, & Pauli, 2005; Fox et al., 2000; Horstmann & Bauland, 2006; Nummenmaa, Hyönä & Calvo 2010; Kutas & Hillyard 1980).

Problems with the existing studies

- These studies do not compare the respective processing of valence and truth-conditional meaning within one and the same word.
- They assume a simplistic view of how information is recruited to specify meaning during language processing.

The Valence Prominence Hypothesis (VPH)

- ‘Prominence’ captures the idea that information is differentially cognitively accessible during language comprehension or production.

→ gradable notion: different features may be more or less prominent.

1. Valence may be more prominent than most features of truth-conditional content, but not necessarily all of them.

2. We might also expect to find differences between the domains of (non-valence) information.

Comparing valence with different truth-conditional domains...

▪ **Concreteness vs. abstractness**

- **long-established as a central dimension of meaning** (Paivio 1971; James 1975; Rudin 1980; Paivio 1986; Kroll et al 1986; Schwanenflugel et al 1988)
- **has a documented interaction with valence** (Kousta et al, 2010; Winter 2020).

▪ **Physical vs. psychological/social**

- **a core domain of reasoning to which infants are attuned** (e.g., Newman et al. 2010; Leslie & Keeble 1987, Spelke et al. 1995; Saxe et al. 2005)
- **continues to be important in adulthood** (e.g., Strickland et al. 2017, Kuhn et al. 2021).

▪ **Animate vs. non animate**

- **acquired early** (Rakison & Poulin-Dubois 2001)
- **has a distinct neurophysiology** (Caramazza & Shelton 1998)
- **is crucial to inferring/learning causation** (e.g., Spelke, Phillips, & Woodward 1995)
- **in word learning** (e.g., Childers & Echols 2004)
- **in mindreading** (e.g., Baron-Cohen 1995)
- **and reasoning about biological processes** (e.g., Carey 1985).
- + **linguistically special: most if not all languages show some grammatical reflexes of animacy** (e.g., Vihman & Nelson 2019).

... and with syntactic category

A good comparison...

> as a different but crucial linguistic dimension.

→ information accessed early upon lexical retrieval

(e.g., Strik Levers Costa XXXX; Frazier & Rayner 1982; Mitchell 1987; Neville et al 1991; Tanenhaus et al 1995; Friederici et al 1996; Friederici 2002).

→ information acquired early before truth-conditional semantic knowledge.

(e.g., Brown 1973; Gleitman 1990; Naigles 1990; Mintz 2003; Gómez & Gerken 1999; Gerken, Wilson & Lewis 2005).

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EXPERIMENTS

Speeded categorization

2 x 1 within-subjects

blocked design (counterbalanced)

- **Task** (**Valence**, **Truth-Conditional**)
- Items presented 3x per block (randomized)
- DM: key board press F or J (counterbalanced)

7 exps

270 Native American English speakers on Prolific, £9/hr, ~15 min median time



progress

mud

F
positive

J
negative



progress

mud

Valence
Task

F

positive

J

negative



progress

mud

Truth-Conditional Task

F

concrete

J

abstract

Predictions: Reaction Time

VFH

Valence < $\forall$ TruthCond

TCFH

$\forall$ TruthCond < Valence

VPH

Valence < $\exists(\wedge \diamond \forall)$ TruthCond

Stimuli

40 words in each exp, within lexical category

10 per 4 possible feature combination

Feature values came from corpus or pre-test norming

Stimuli from corpus/norms

- **Valence**: Warriner et al 2013
- **Conc. vs. Abs.**: Brysbaert et al 2013
- **Phys. vs. Psych./Soc.**: 2 pre-test norming studies conducted
- **Animacy**: Wordnet synsets with 'living organism' or 'alive' as superset (then hand-selected)
- **Syntax**: unambiguous nouns vs. verbs

RT Analysis

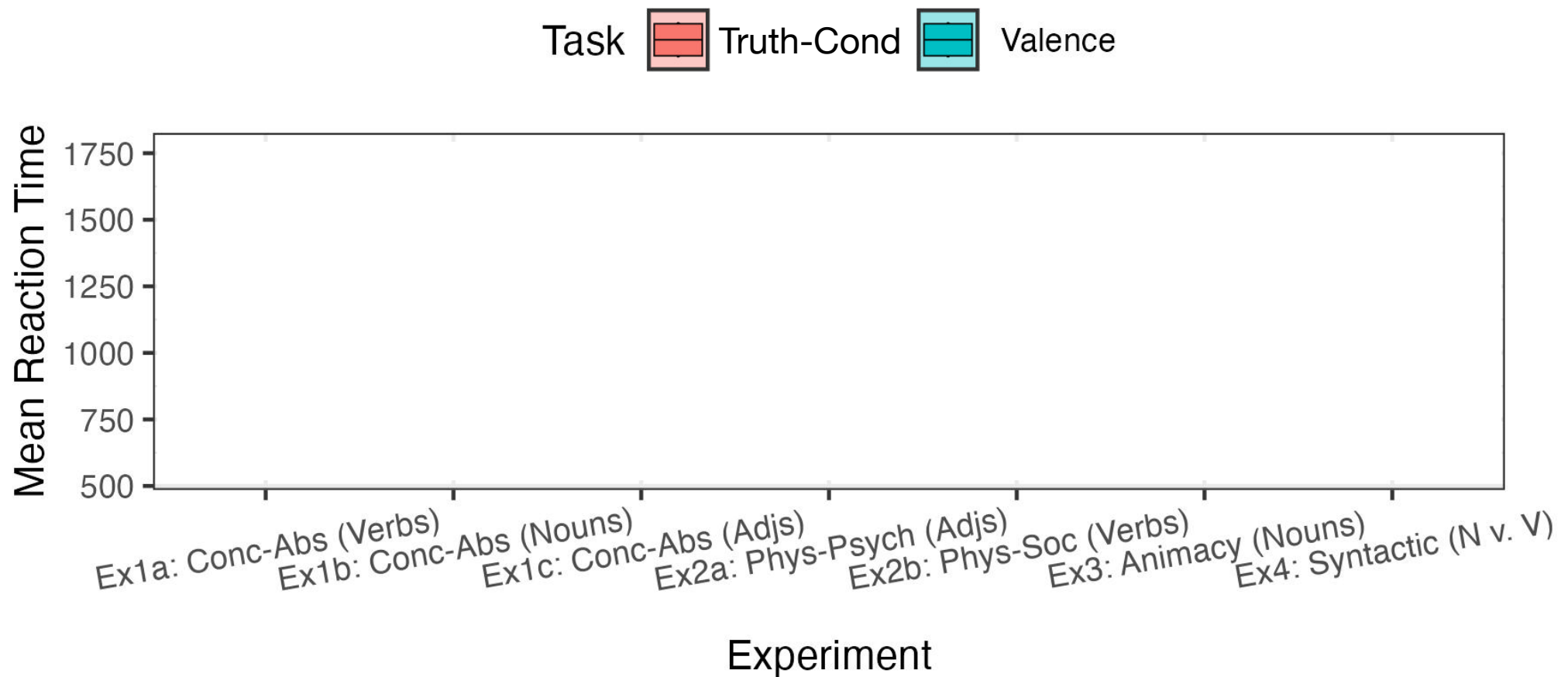
- Inaccurate participants removed ($\text{Acc} < 75\%$)
- Only Accurate trials included
- Excluded trials where $\text{LogRT} > 3 \cdot \text{IQR}$

Range of data excluded across exps: ~10-15%

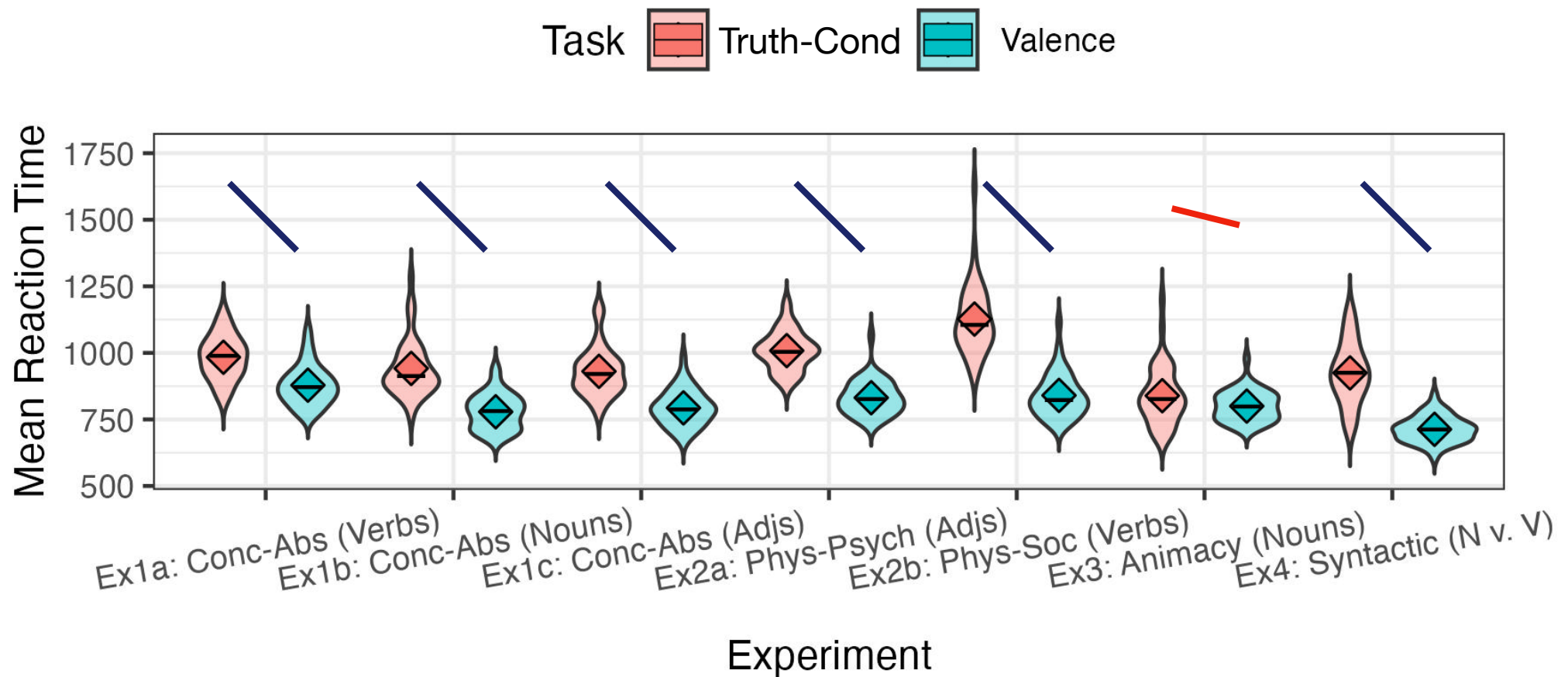
Model:

$\text{LogRT} \sim \text{Task} + (1 + \text{Task} \mid \text{ID}) + (1 + \text{Task} \mid \text{Word})$

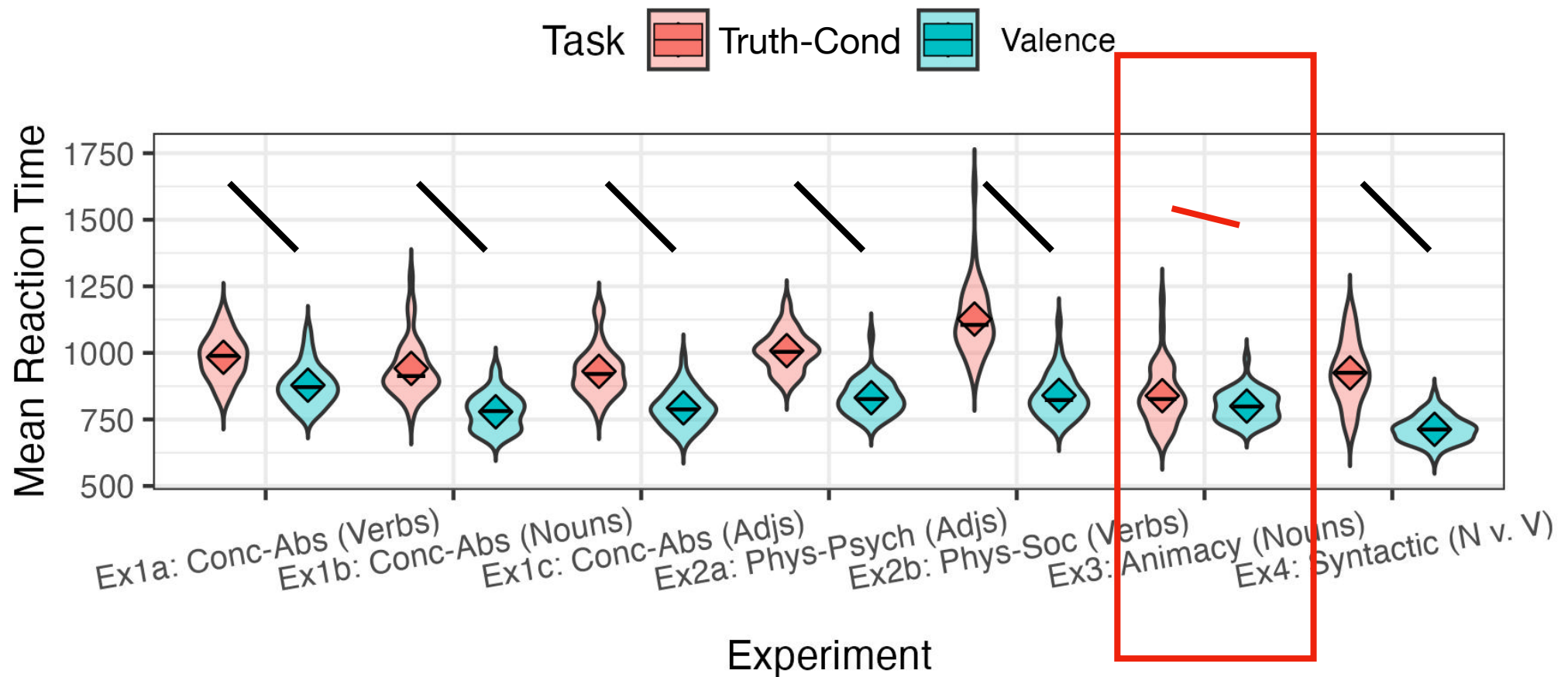
Results: Categorization



Results: Categorization

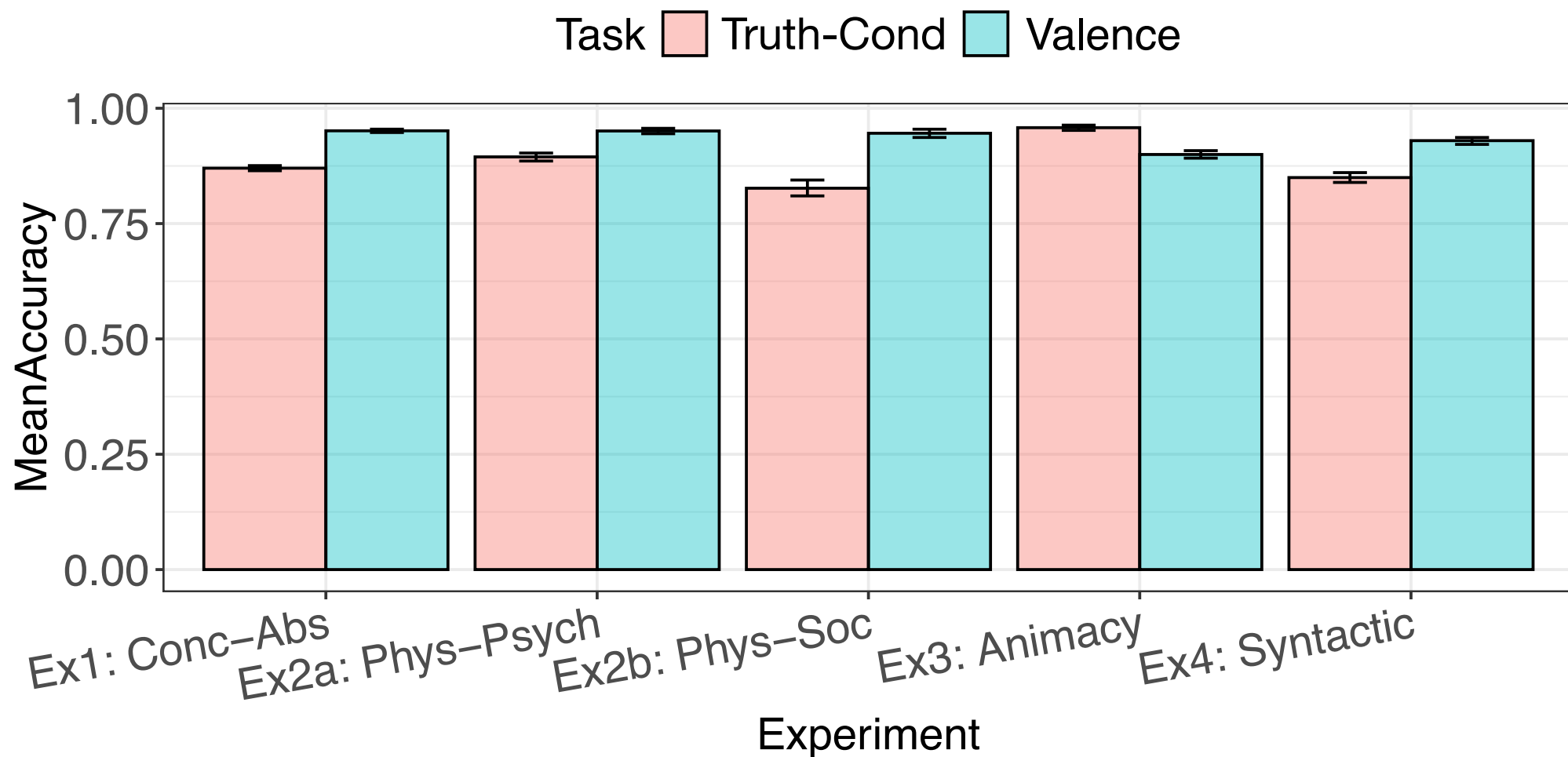


Results: Categorization



**Not just uncertainty about
conceptual meaning**

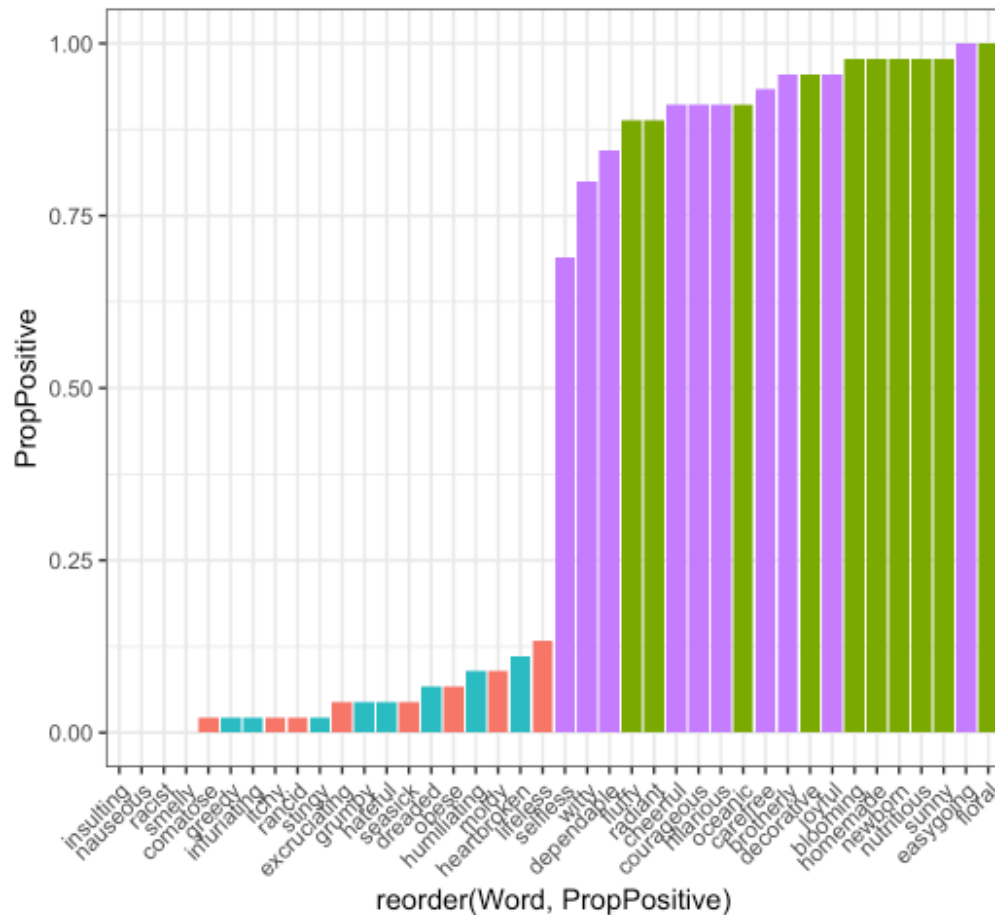
Total Accuracy



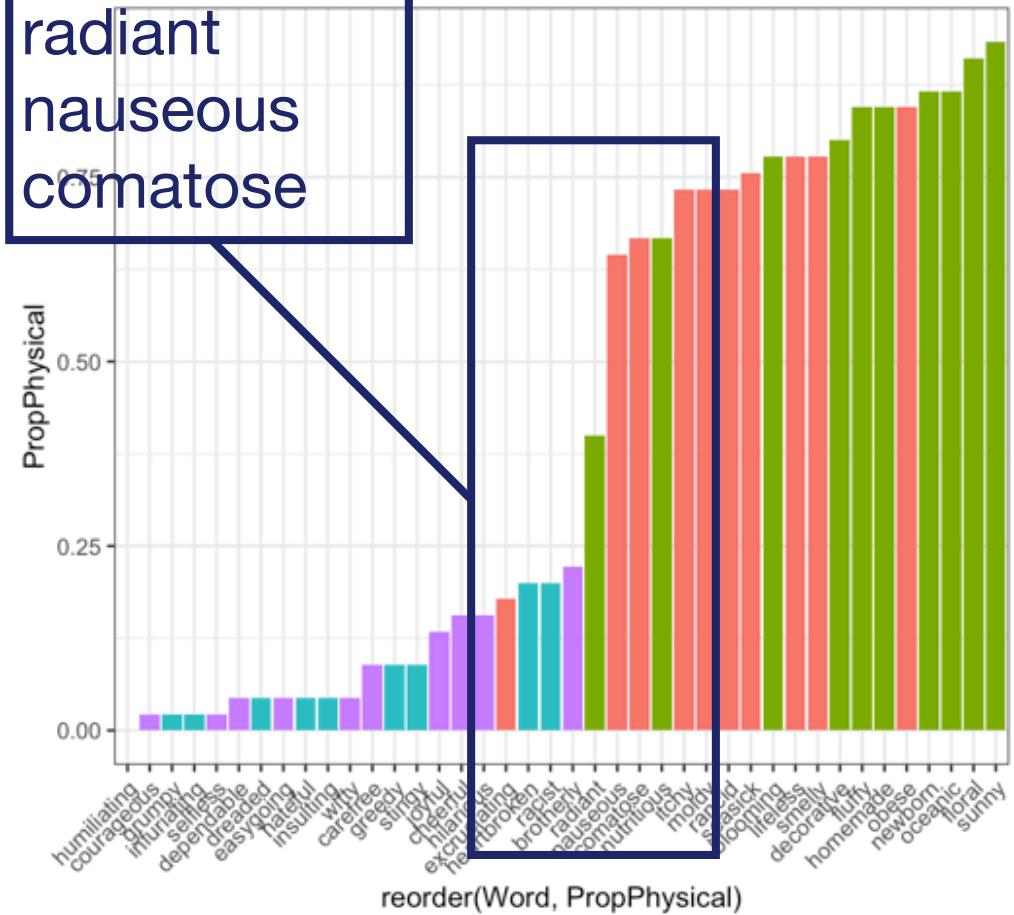
Responses Exp2b

Feature Combo

- negative-Nouns
- negative-Verb
- positive-Nouns
- positive-Verb



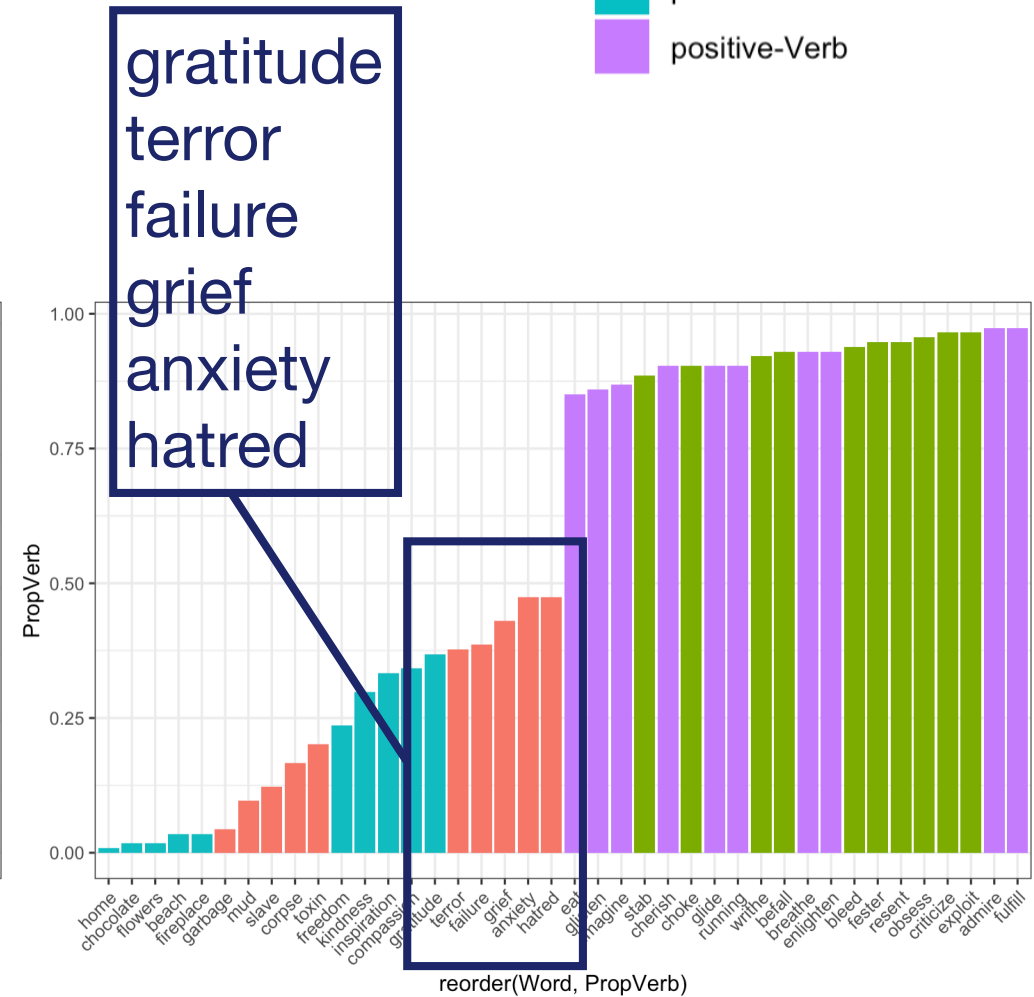
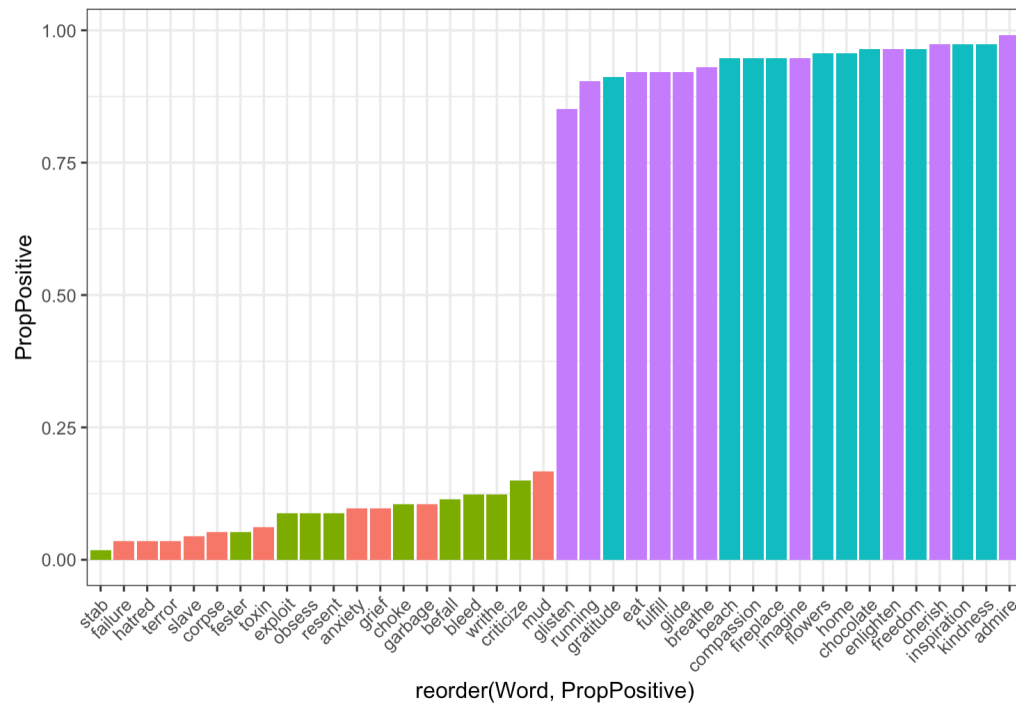
excruciating
radiant
nauseous
comatose



Responses Exp4

Feature Combo

- negative-Nouns
- negative-Verb
- positive-Nouns
- positive-Verb



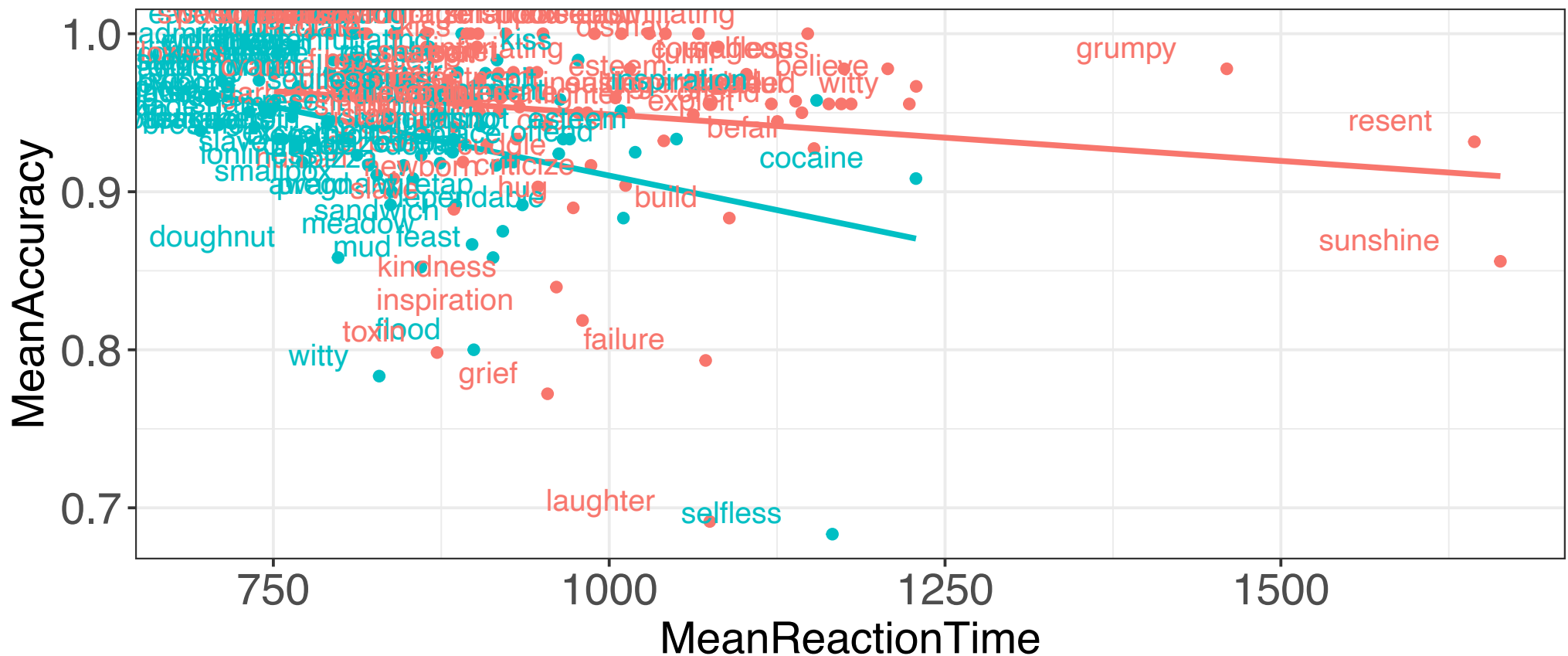
Analysis: Accuracy Control

For each study (n=7), take the 10 words where Accuracy on **Truth-Cond.** task $\geq$ Accuracy on **Valence**

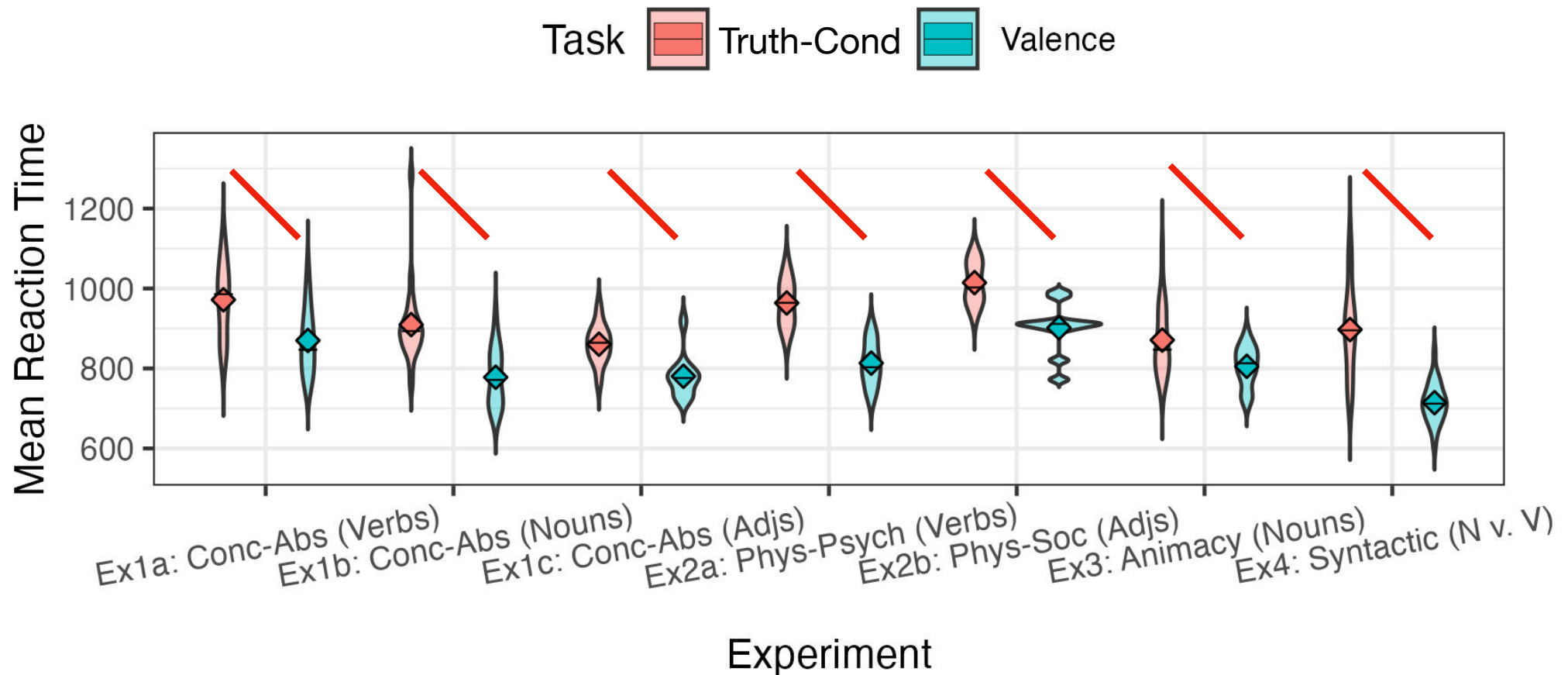
$$\text{LogRT} \sim \text{Expe} * \text{Task} + (1 + \text{Task} \mid \text{ID}) + (1 + \text{Task} \mid \text{Word})$$

ACC_{Truth-Cond} ≥ ACC_{Valence}

Task  Truth-Cond  Valence



RT for Most Acc. Words



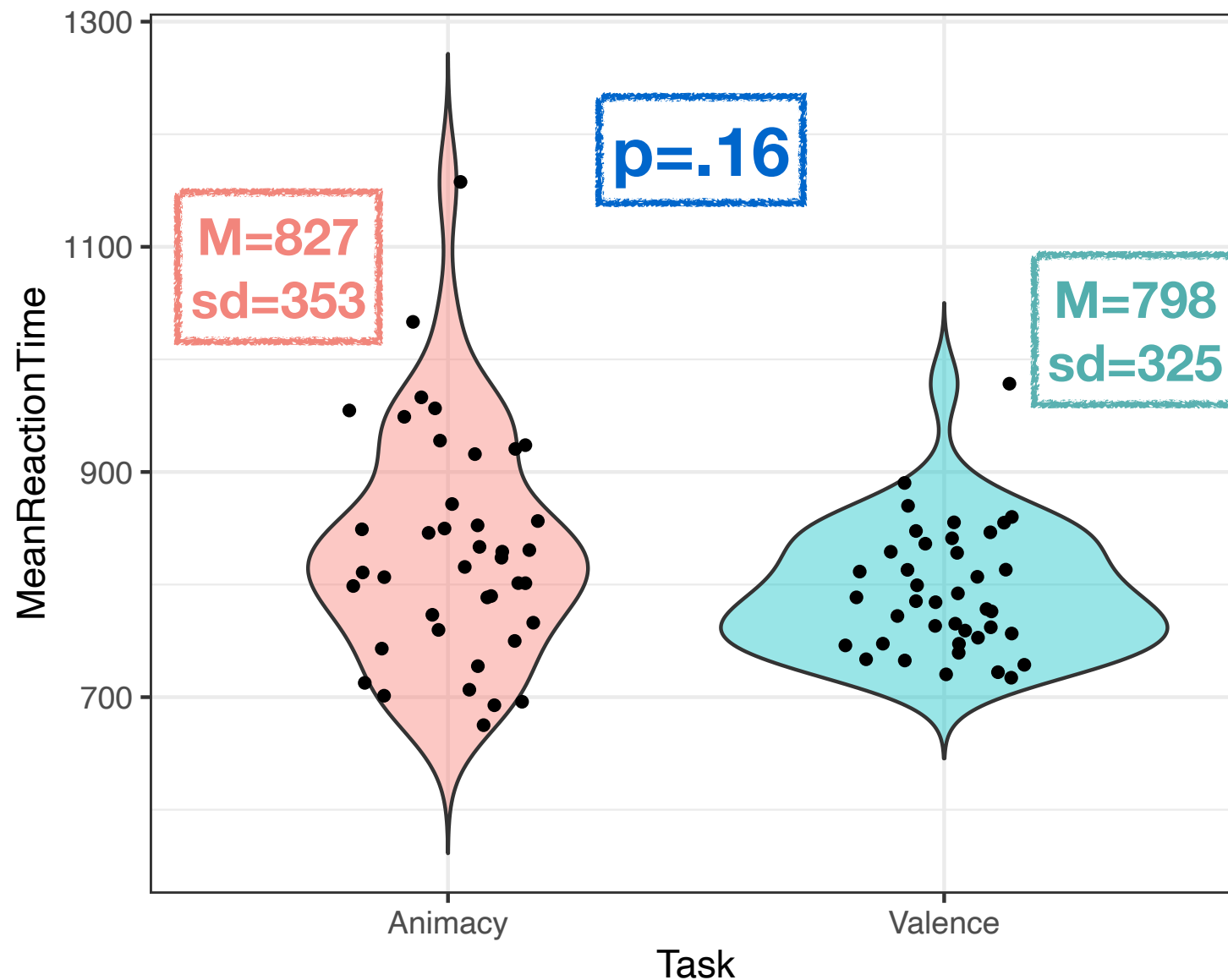
**Not just uncertainty about
truth-conditional meaning**

Not just Habit

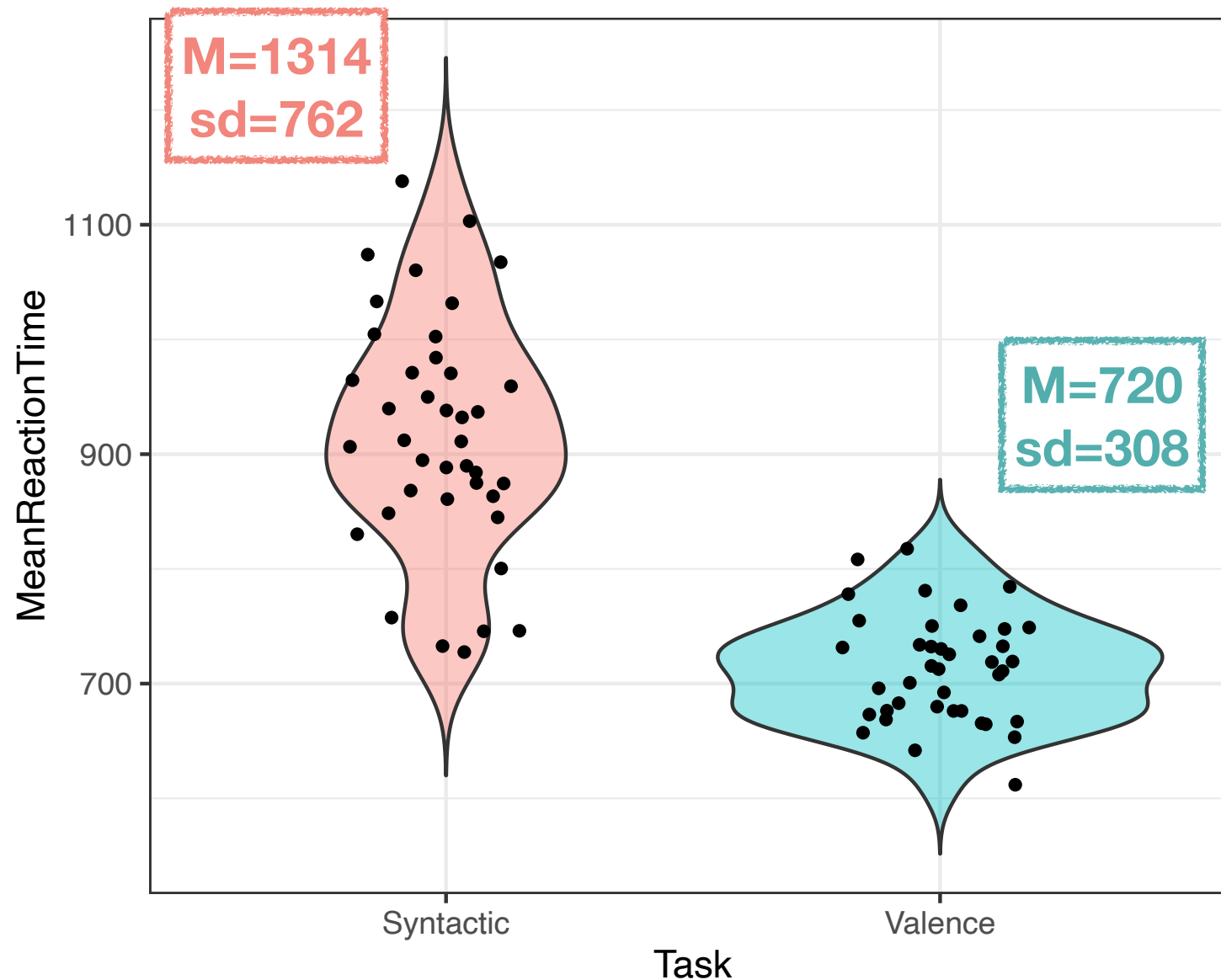
Not just uncertainty about
truth-conditional meaning

Not just Habit

We aren't in the habit of explicitly judging Animacy



But we do for Syntactic category when we deploy language



Is affective info more
prominent for determining
word similarity in an **implicit**
task?

progress

push
pull

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

Similarity judgements

Implicit measure of word meaning

(Osgood et al 1957; Degen 2015; De Deyne, Perfors & Navarro 2016, a.o.)

4 x 1: Feature Match

DM: slider from 0 (dissimilar) to 100 (similar)

MaxMismatch, DissimilarControl (Antonyms)

MaxMatch, SimilarControl (Synonyms)

**Matching
ValenceOnly**

vs.

**Matching
TruthCondOnly**

progress

(**tc** , **val**)

push

(**tc** , **val**)

pull

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

progress

(**abs** , pos)

love

(**abs** , neg)

hate

Matching
TruthCondOnly

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

progress

(abs , pos)

love

(con , pos)

hug

**Matching
ValenceOnly**

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

Predictions: Similarity

VFH

$\text{ValMatch} > \forall \text{TCMatch}$

TCFH

$\forall \text{TCMatch} > \text{ValMatch}$

VPH

$\text{ValMatch} > \exists (\wedge \diamond \forall) \text{TCMatch}$

Stimuli

Words from categorization experiments (7 lists)

All possible combinations, within-list

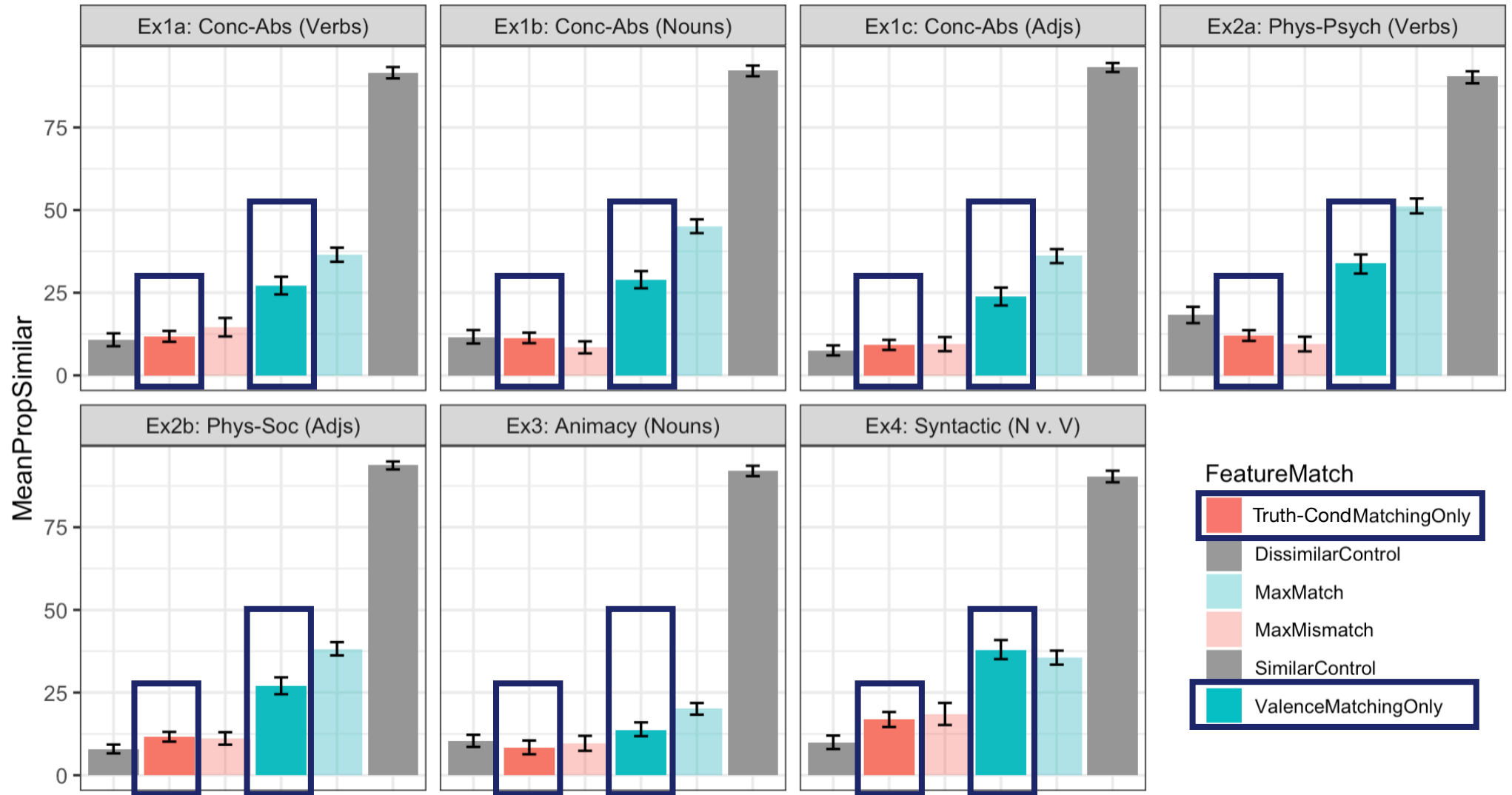
- Unordered
- Same-word pairs removed

60 items

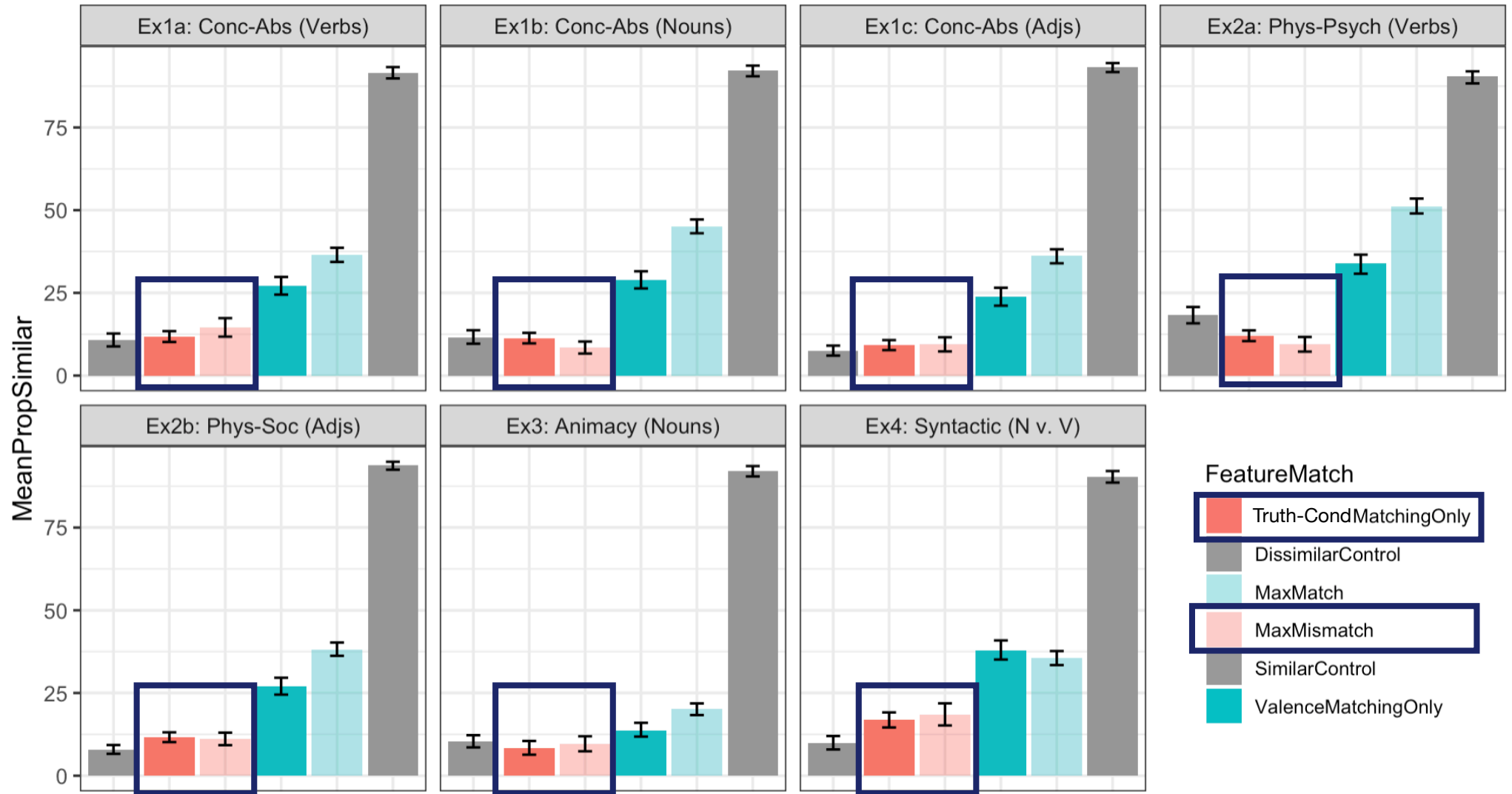
- 10 randomly sampled from the 4 FeatureMatch Conditions
- +10 Similar Controls (Synonyms),
- +10 Dissimilar controls (Antonyms)

Rating $\sim$ FeatureMatch + (1 + FeatureMatch | ID) + (1 | Word)

Results (n=350)



Results (n=350)



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DISCUSSION

Summary

The results of the two sets of experiments are strikingly consistent with each other.

- **Valence information** is overall **more prominent than truth-conditional information** in lexical processing:

First set of studies:

valence categorization is faster than most categorizations of truth-conditional meaning.

Second set of studies:

valence features drive similarity, while truth-conditional features seemed to have almost no impact.

- **Animacy** behaves differently from the other foils, **patterning closest to valence**.

⇒ Support for the VPH, not for the TCFH or for the VFH.

Exploring prominence

- Here, prominence is used as a pre-theoretic notion capturing the differential cognitive accessibility of information during comprehension.
- In psycholinguistics, prominence (/ “salience”) is typically treated as graded accessibility, though theories differ in the mechanisms responsible.
 - > **Structural approaches** link prominence to grammatical and discourse organization, ranking entities such as subjects or topics higher in discourse representation (Grosz, Joshi, & Weinstein, 1995).
 - > **Lexical approaches** treat salience as a property of lexical representations shaped by frequency, familiarity, and conventionality (Giora, 1997).
 - > **Memory-based accounts** attribute salience to activation dynamics in working memory, with accessibility modulated by recency, frequency, and retrieval interference (Chafe, 1994; Ariel, 1990; Gibson, 1998; Lewis & Vasishth, 2005).
 - > **Other work emphasizes attention, prediction, and event structure as additional sources of salience** (Altmann & Kamide, 1999; Levy, 2008; Zwaan & Radvansky, 1998).
- ⇒ Does the ordering of lexical features observed here remains stable across contexts?

